

Bayesian Learning for Individualized Treatment Decisions in Personalized Medicine

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Motivation

- Not all participants will react equally to an intervention
- Characterizing this heterogeneity in intervention effects is key to improving patients outcomes

The predicted individual treatment effects (PITE) framework

- PITE estimates individual treatment effects.
- PITE consists of the difference between experimental(E) and control (C) **prediction** for each individual (Jaki et al. 2024).

$$\text{PITE}_i = f_E(X_i) - f_C(X_i), \quad f(\cdot) \text{ is a predictor}$$

! Challenge

- PITE is unobserved.
- The best method to estimate the PITE is not necessarily the best fitted model.

Modeling Strategies

Solano et. al (2026)

	Heterogeneity	Linear	Non_Linear	Overfitting	Regularization	Tree_Regression
The lasso (lasso)						
The Bayesian Lasso (blasso)						
Spike and Slab Regression (spikeslab)						
Ridge (ridge)						
Random Forest (rf)						
Quantile Random Forest (qrf)						
Projection Pursuit Regression (ppr)						
Principal Component Analysis (pcr)						
Polynomial Kernel Regularized Least Squares (krlsPoly)						
Penalized Linear Regression (penalized)						
Partial Least Square (pls)						
Non-Negative Least Squares (nnls)						
lm with leapBackward (leapBackward)						
lm with Forward (leapForward)						
lm Stepwise (leapSeq)						
Linear Regression with Stepwise Selection (lmStepAIC)						
Linear Regression (lm)						
GLM with Stepwise Feature Selection (glmStepAIC)						
Generalized Linear Model (glm)						
Elasticnet (enet)						
Cubist (cubist)						
CART (rpart2)						
Boosted Tree-shrinkage (bstTree)						
Boosted Tree-Boost number (blackboost)						
Boosted Linear Model (BstLm)						
Boosted Generalized Linear Model (glmboost)						
Bayesian Ridge Regression Model Averaged (blassoAveraged)						
Bayesian Ridge Regression (bridge)						
Bayesian Regularized Neural Networks (brnn)						
Bayesian Generalized Linear Model (bayesglm)						
Bayesian Additive Regression Trees (bartMachine)						

Bayesian Learning Strategies for PITE

Solano et. al (2026)

Dimensionality

Bayesian Ridge Regression ("bridge"), Spike and Slab Regression ("spikeslab"), The Bayesian Lasso ("blasso").

Complex outcomes

Bayesian Generalized Linear Model ("bayesglm"), Bayesian Regularized Neural Networks ("brnn").

Regression Trees

Bayesian Additive Regression Trees ("bartMachine") (Lamont et al. 2016)

Missing Information and PITE

$$\tau(X_i) = f_1(X_i) - f_0(X_i)$$

- PITE depends on the covariate information available for each patient.

Under an appropriate imputation model [Rubin, 1987]

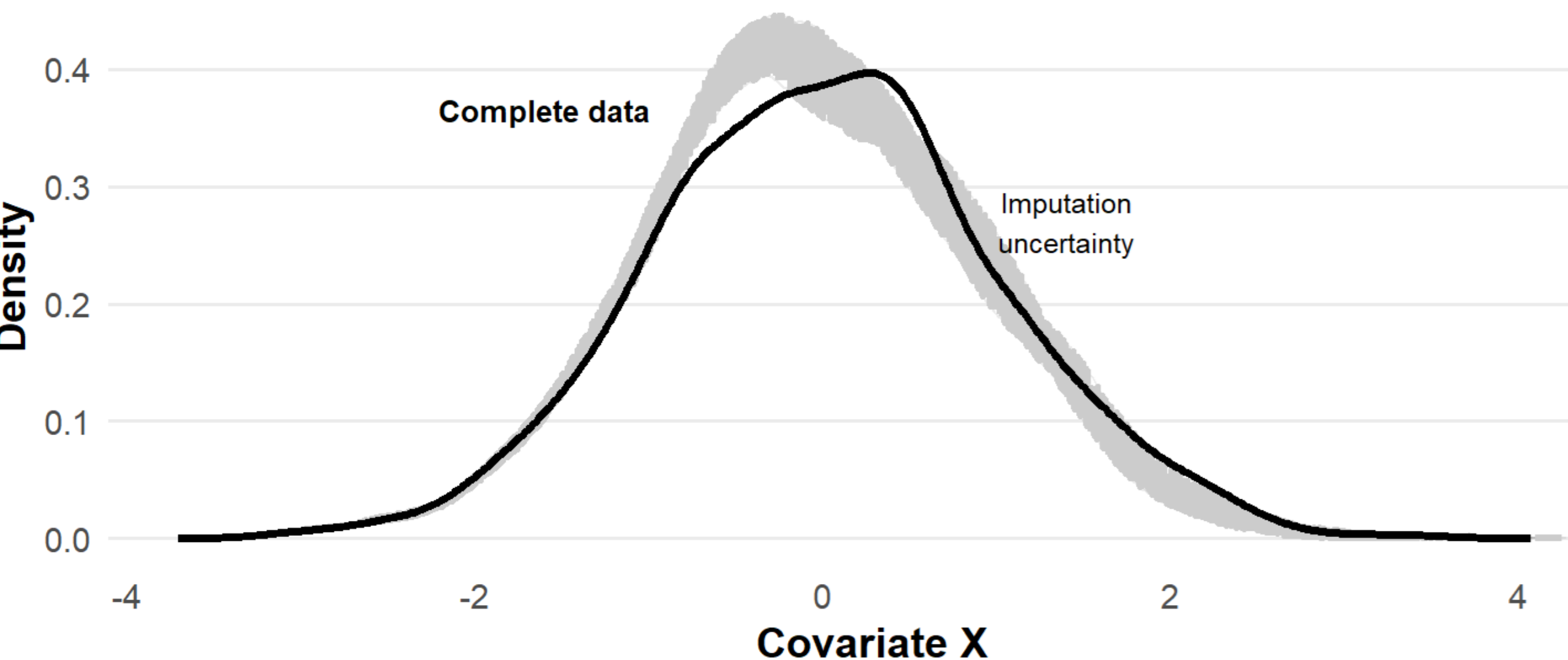
- MI is specifically designed to propagate missing-data uncertainty.

Imputation's vulnerability [Sterne et al., 2009]

- Based on assumptions that do not hold.
- Imputation model can be misspecified

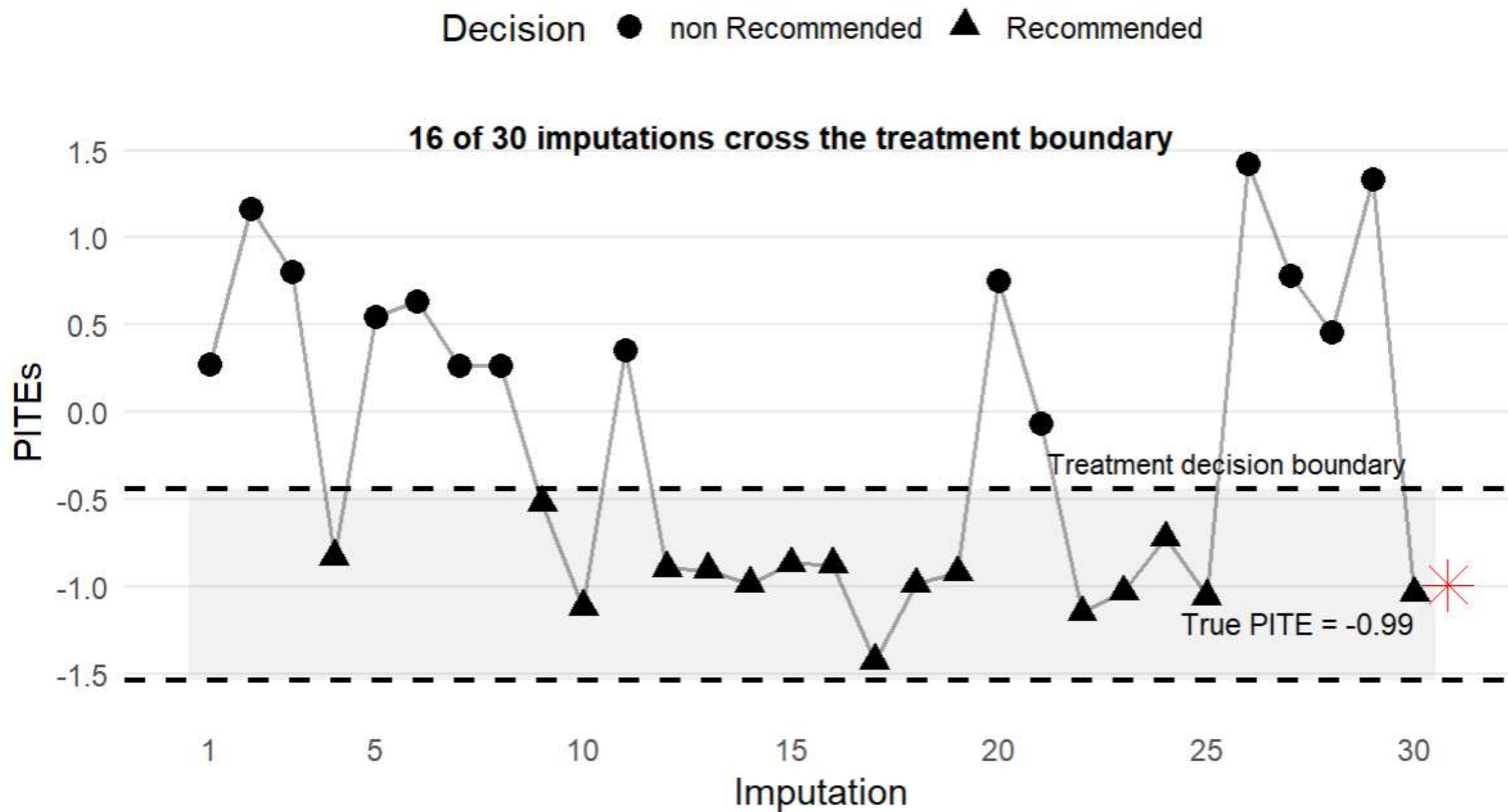
Imputation changes the predictor distribution learned by PITE

Black = complete data | Grey = 30 imputed datasets | band = 10th–90th percentile across imputations



Same patient. Different imputation. Different treatment decision

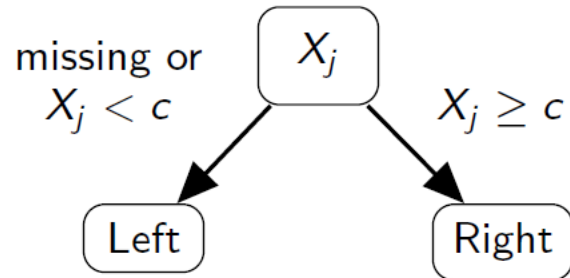
Patient 15 | Mean PITE = -0.21 | SD = 0.87



Free-of-imputation BART

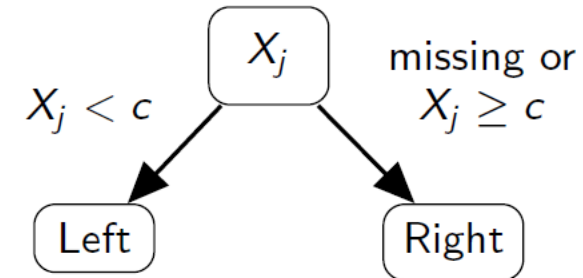
Rule 1

Missing $\rightarrow$ Left



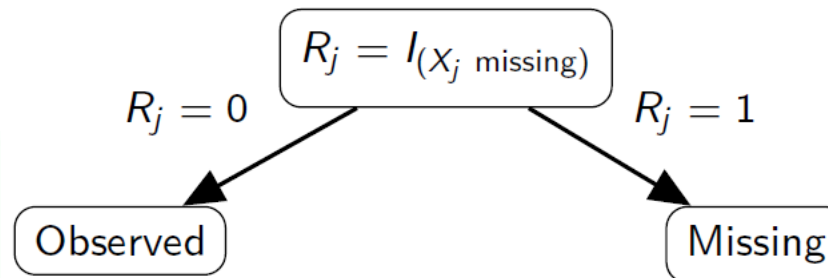
Rule 2

Missing $\rightarrow$ Right



Rule 3

Missingness indicator



Key point

How can we estimate trustworthy, robust PITE ?

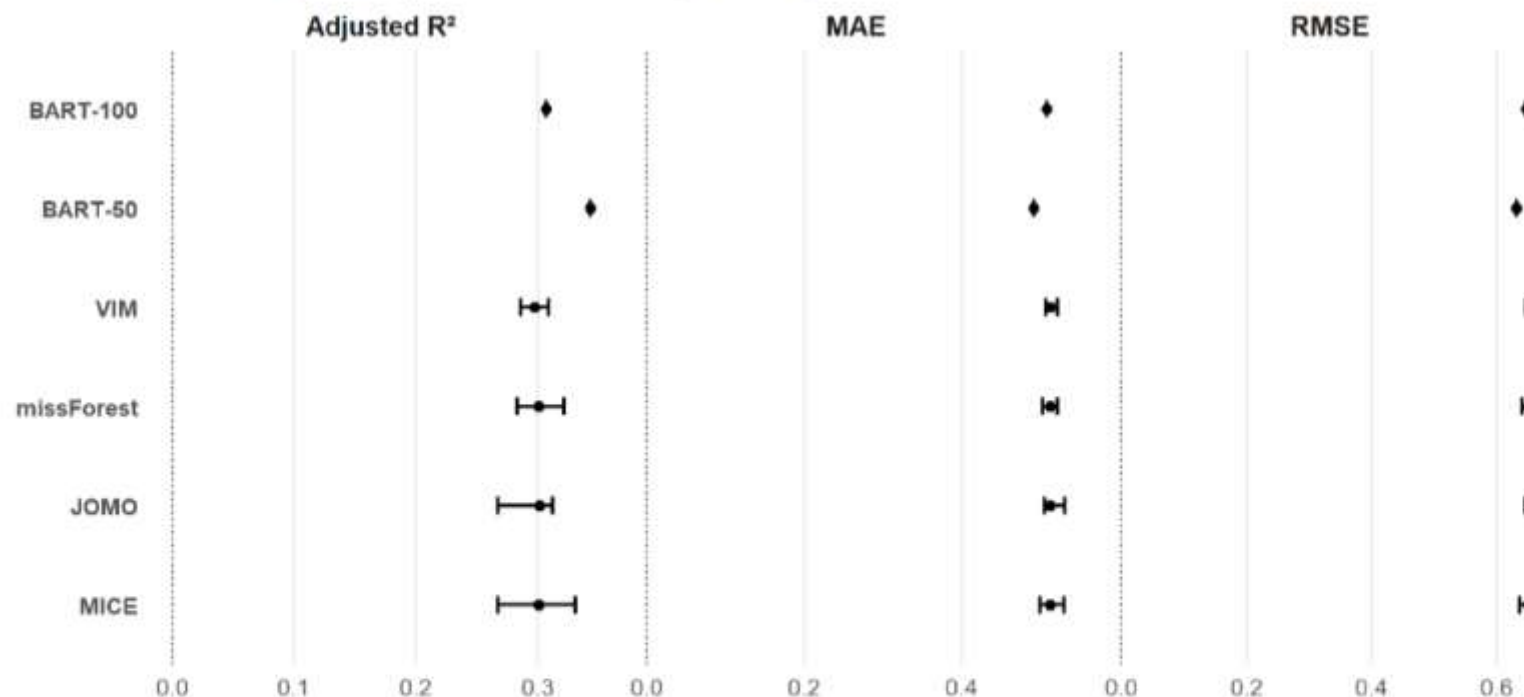
Application

21 of 32 covariates contain missing information (65.6%)

Patients by treatment group: $n_t \approx 240, t = 1, \dots, 4$

Predictive performance with and without imputation

Points = mean prediction performance; intervals = range across imputations



Computational burden: for $m = 10$ the imputation time is approximately:

- MICE: 5.60 min
- missForest: 5.51 min
- VIM: 1.27 min
- JOMO: 15.14 hours

Conclusion

- Learning missingness yields lower or equivalent risk than imputation
- Free-missingness is a robust, efficient, and conceptually simple alternative for PITE
- Free-of-imputation BART preserves information
- Tighter uncertainty, computationally efficient
- Robust across heterogeneous clinical settings (MAR)

Thank you

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